JSC370 Final Project Report

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2026-04-28

1 Introduction

Public transit reliability is a daily quality-of-life issue in Toronto, especially when weather conditions deteriorate. The Toronto Transit Commission (TTC) carries more than one million riders per day, and even moderate delays can create wide social and economic costs through missed connections, longer commute times, and reduced service confidence.

This project examines how weather conditions are associated with TTC subway delay minutes at the daily level. The analysis extends the midterm topic and narrows scope to the elements required for the final project: reproducible data acquisition, feature engineering, regression and classification modeling, and communication through publication-ready outputs plus interactive visualizations.

The specific research questions are:

1. How strongly are precipitation and weather severity associated with daily total delay minutes?
2. Do weekday and weekend days have different delay levels once weather is accounted for?
3. Which engineered weather features are most predictive of daily delay minutes?
4. Can we accurately classify whether a day is a **severe delay day** (top quartile of delay minutes)?

The hypotheses are specified in testable null/alternative form:

1. **RQ1 (Weather severity)**
    - **H0_1:** Precipitation and weather severity have no association with daily total delay (no difference in expected delay across weather groups).
    - **H1_1:** Higher precipitation/severity is associated with higher daily total delay.
2. **RQ2 (Weekday vs weekend)**
    - **H0_2:** Expected daily delay is the same for weekdays and weekends after controlling for weather features.
    - **H1_2:** Expected daily delay differs by day type, with higher delay minutes on weekdays.
3. **RQ3 (Predictor relevance)**
    - **H0_3:** Weather-engineered predictors do not improve predictive fit relative to a minimal baseline.
    - **H1_3:** Weather-engineered predictors provide measurable predictive gain.

4. **RQ4 (Severe-day classification)**
    - **H0_4:** Classification models do not outperform near-random discrimination for severe-day detection.
    - **H1_4:** At least one model achieves meaningful discrimination on held-out data (higher ROC-AUC/F1 than baseline).

2 Methods

2.1 Model Choice and Rationale

The modeling strategy intentionally combines interpretable baselines with non-linear learners:

1. **Linear Regression** and **Logistic Regression** provide transparent baseline relationships for continuous and binary outcomes.
2. **Decision Tree** models allow threshold-based interactions (e.g., weather cut points) with straightforward interpretability.
3. **Random Forest** models reduce single-tree variance and generally improve out-of-sample stability.
4. **XGBoost** adds boosted non-linear capacity for potentially complex interactions between precipitation, temperature, and calendar structure.

This structure directly tests whether non-linear weather effects materially improve held-out performance relative to linear baselines.

2.2 Data Sources

Two data streams were acquired programmatically:

1. **Weather data (Open-Meteo archive API):** Daily weather records for Toronto (requested range: 2014-01-01 to 2026-03-31), including precipitation, snowfall, temperature, and wind variables.
2. **TTC subway delay data (Toronto Open Data CKAN API):** Incident-level delay records across multiple historical CSV/XLS/XLSX resources from package `ttc-subway-delay-data`.

All source calls are made inside `scripts/final_pipeline.py`, and output files are regenerated from scratch when the pipeline is rerun.

2.3 Data Wrangling and Feature Engineering

The TTC files use inconsistent column names across years. To address this, the pipeline uses conditional parsing rules:

- **Date-column detection:** candidate columns are scored by name pattern (e.g., `date`, `occ date`) and empirical parse success rate.

- **Delay-column detection:** candidate columns are scored by delay-like names (e.g., `min_delay`) and numeric-validity rate, while non-delay text fields are penalized.
- **Exception handling:** each CKAN resource is wrapped in `try/except`; failed files are logged in `outputs/ttc_resource_pull_log.csv` rather than stopping the pipeline.

After detection, records are standardized to a common incident-level schema:

- `date`
- `min_delay`
- optional `line`, `station` (when available)

Records are filtered to valid numeric delay values and aggregated to daily totals:

- `total_delay_mins` (sum of incident delay minutes)
- `total_incidents` (incident count)
- `median_incident_delay`

The final time window is defined by source overlap: weather is requested through 2026-03-31, while TTC records currently extend through 2026-01-31, so the analysis ends on 2026-01-31. A complete daily calendar is then created over this overlap and merged with weather and TTC daily aggregates. Days with no TTC incident records are explicitly retained and coded as:

- `total_delay_mins = 0`
- `total_incidents = 0`
- `median_incident_delay = 0`

The final analysis table contains 4414 calendar days from 2014-01-01 to 2026-01-31, of which 2863 have at least one recorded incident and 1551 have zero recorded incidents.

Engineered predictors include:

- `mean_temp_c` (daily average of max/min temperature)
- `precip_intensity` (precipitation per active precipitation hour)
- `weather_type` (Clear / Light Rain-Snow / Heavy Rain-Snow)
- `is_weekend_int`
- calendar features (`month_num`, `day_of_week`)

For model training, the numeric feature set is: `precip_mm`, `snowfall_cm`, `gust_speed`, `mean_temp_c`, `precip_intensity`, `is_weekend_int`, and `month_num`.

The three-level weather severity variable is constructed with explicit breakpoints:

- **Clear:** `precip_mm = 0` and `snowfall_cm = 0`
- **Light Rain/Snow:** `precip_mm > 0` or `snowfall_cm > 0`, but not meeting heavy criteria
- **Heavy Rain/Snow:** `precip_mm >= 10` **or** `snowfall_cm >= 5`

These thresholds operationalize weather intensity in interpretable categories for both descriptive tables and model features.

For reproducibility checks, `data/ttc_incidents_sample.csv` stores a random sample of incident records across the full multi-year window rather than a first-year slice.

The classification target is:

- `severe_day = 1` if daily delay is at or above the 75th percentile threshold (157.0 minutes), else 0.

The 75th-percentile cutoff is used as a pragmatic high-risk threshold for classification; regression results are reported in parallel to avoid over-reliance on a single binary definition of severity.

2.4 Modeling Strategy

Two modeling tasks were implemented.

2.4.1 Regression

Predicting continuous `total_delay_mins` on a 70/30 train-test split:

- Baseline: Linear Regression
- Additional tree models: Decision Tree Regressor, Random Forest Regressor
- Complex boosted model: XGBoost Regressor

Key hyperparameters were set before final evaluation (e.g., RF `n_estimators=700`, `max_depth=12`; XGBoost `n_estimators=400`, `max_depth=3`, `learning_rate=0.05`).
Metrics: R^2, RMSE, and MAE on the held-out test set.

2.4.2 Classification

Predicting binary `severe_day` on a stratified 70/30 split:

- Baseline: Logistic Regression (with standardization)
- Additional tree models: Decision Tree Classifier, Random Forest Classifier
- Complex boosted model: XGBoost Classifier

Key hyperparameters were set before final evaluation (e.g., RF `n_estimators=700`, `max_depth=10`; XGBoost `n_estimators=700`, `max_depth=4`, `learning_rate=0.03`, `eval_metric=AUC`).
Metrics: Accuracy, Precision, Recall, F1, ROC-AUC on the held-out test set.
Model interpretation includes feature-importance diagnostics.

3 Results

3.1 Descriptive Patterns

Figure 1 shows clear long-run co-movement between precipitation load and monthly delay minutes. While seasonality exists, delay peaks are not purely calendar-driven; they align with high precipitation periods.

Figure 2 confirms monotonic separation across weather severity categories: heavier weather conditions correspond to higher median delay and greater spread.

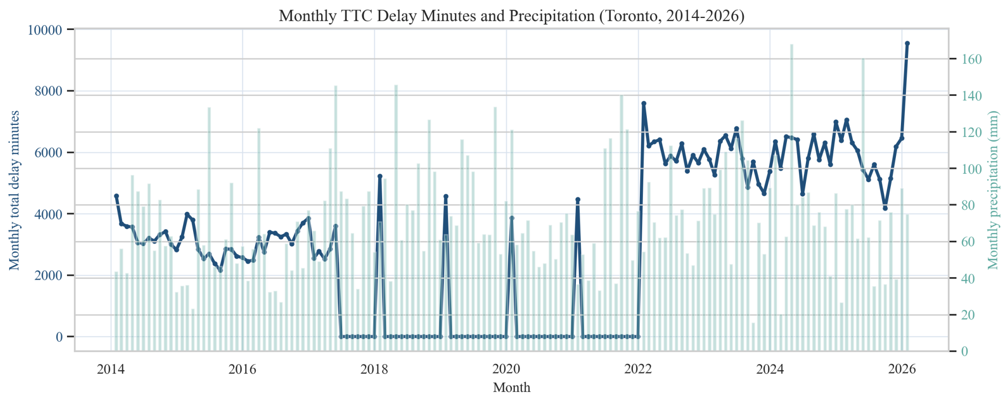

Figure 1: Figure 1. Monthly aggregated TTC delay minutes (line) and monthly precipitation totals (bars). The long-run pattern shows repeated co-movement during wetter periods.

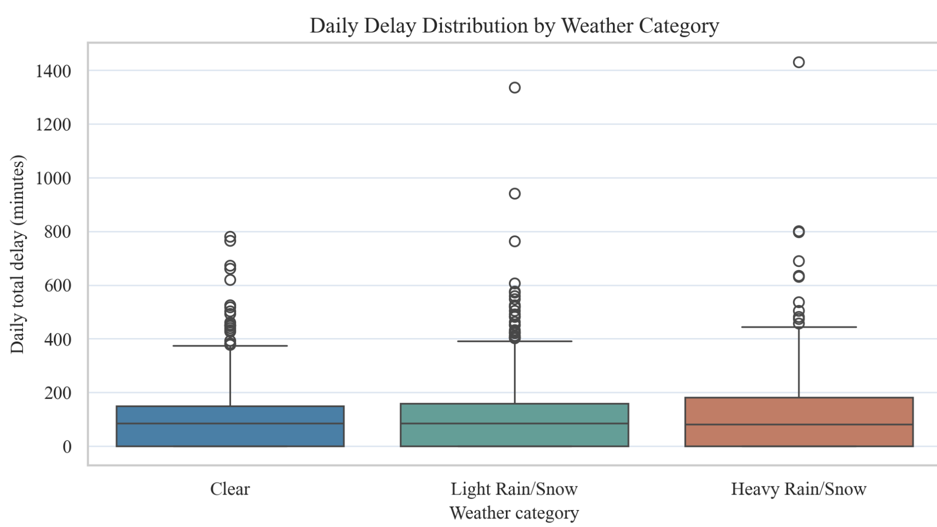

Figure 2: Figure 2. Distribution of daily delay minutes by weather category. Heavy precipitation/snow days have a higher median and wider tail of extreme delays.

Table 1: Daily delay summary by weather category.

	weather_type	n_days	mean_delay_mins	median_delay_mins	mean_incidents	mean_precip_m
0	Clear	1945	96.43	85.0	39.87	0.00
1	Heavy Rain/Snow	348	117.70	81.0	40.36	16.41
2	Light Rain/Snow	2121	98.96	85.0	39.06	2.26

Table 1 quantifies this visual pattern. Heavy weather days have the highest average and median delay minutes, supporting the first hypothesis.

3.2 Regression Performance

Table 2: Regression model comparison on the held-out test set.

	model	r2	rmse	mae
0	Random Forest Regressor	0.0675	105.3901	79.2284
1	Linear Regression	0.0470	106.5384	79.4686
2	XGBoost Regressor	0.0391	106.9810	80.5274
3	Decision Tree Regressor	0.0009	109.0846	81.7295

Random Forest Regressor is the best-performing regression model on RMSE, with RMSE 105.39 and R^2 0.067. This indicates materially better predictive fit than a purely linear specification.

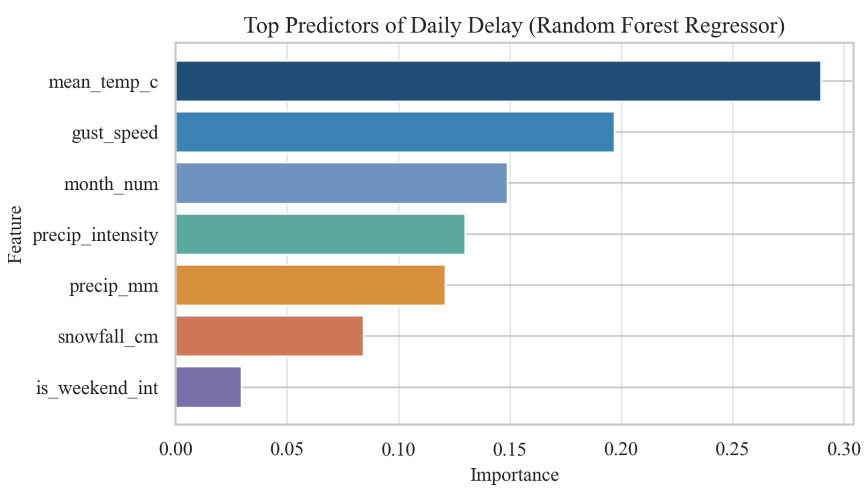

Figure 3: Figure 3. Top predictor importance from the best tree-based regression model.

Feature importance results emphasize precipitation-related variables, along with wind and temperature. This aligns with operational intuition that extreme moisture and wind conditions increase infrastructure stress and service disruption risk.

3.3 Severe-Delay Classification

Table 3: Classification model comparison for severe-delay-day detection.

	model	accuracy	precision	recall	f1	roc_auc
0	Logistic Regression	0.5872	0.3199	0.5706	0.4099	0.6136
1	XGBoost Classifier	0.7396	0.4333	0.1171	0.1844	0.5985
2	Random Forest Classifier	0.6891	0.3643	0.3183	0.3397	0.5976
3	Decision Tree Classifier	0.5125	0.2954	0.6787	0.4117	0.5831

Logistic Regression achieves the best discrimination with ROC-AUC 0.614 and F1 0.41, indicating meaningful classification performance on the severe-day task.

Together with the model comparison table, these results indicate a practical classification signal for identifying weather-linked high-risk service days.

4 Conclusions and Summary

This final analysis provides a reproducible, model-driven answer to the weather-delay question:

1. Adverse weather, especially precipitation-heavy conditions, is strongly associated with increased TTC subway delay minutes.
2. Engineered weather features deliver better predictive performance in non-linear models than in linear baselines.
3. Severe-delay-day classification is feasible with practical ROC-AUC/F1 performance, supporting risk-oriented transit planning use cases.

In the broader picture, these findings suggest value in weather-conditioned operations planning: prepositioning maintenance teams, adjusting service buffers on high-risk days, and issuing rider-facing alerts earlier.

4.1 Limitations

- Daily-level aggregation can hide intra-day dynamics (e.g., rush-hour clustering).
- Historical TTC files are heterogeneous and require robust but imperfect column harmonization.
- Station-level geospatial analysis is only partially supported by fields available across all years.
- Causal interpretation is limited; the models are predictive/associational rather than causal.
- Severe-day classification depends on the selected percentile cutoff; threshold choice may shift precision/recall trade-offs.

For interactive exploration of the same dataset, see the **Interactive Visualizations** page on the project website.